

woodWOP- FILE-Description (MPR-Format)

Valid from woodWOP Version 5.0.775.999

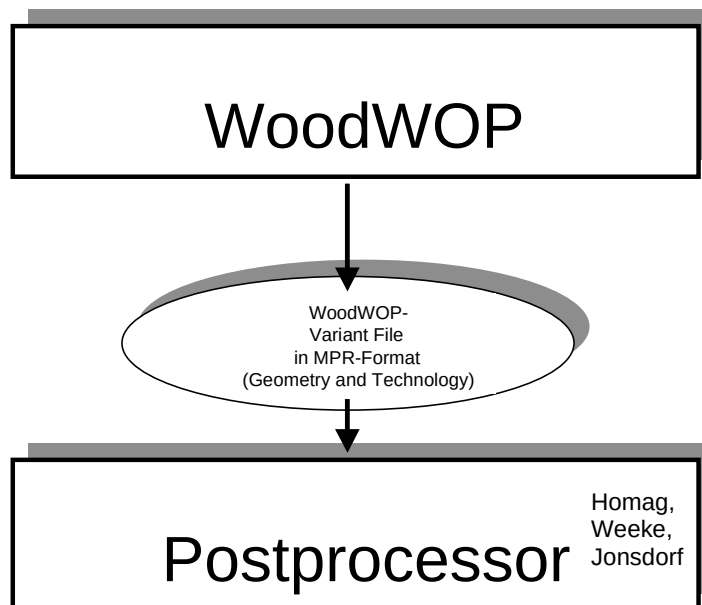


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1 Introduction

The programing system WoodWop generates for the saving of the given data and for introduction to the post processor a so called „Variant File“ in MPR-format. This File for CNC program generating contains the whole work piece description. Out of this variant file you can generate a CNC-program.

The individual sections in the variant file will be devided with a section header, that means, in each section is only one header. The variant file is in ASCII-code.

The variant file is arranged in several blocks:

- **Data head**
Determines the version
- **Variable table**
- **Definition and value assignment of parameters** (variables)
- **Cartesian coordinate systems**
Description of additional defined Cartesian coordinate systems.
- **Contour elements**
Description of single points or elements of the work piece.
- **Processing**
Description of the raw work piece and the single processing steps that have to be done at the work piece.

The sequence of the blocks is fixed and can't be changed. The variable table and the contour elements are optional and are applicable.

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2 Descriptor Of Variant File

The variant file is section oriented . Each section begins with a descriptor which defines the sense of the following argument.

General:

\ = Comment descriptor. Characters between comment characters and row end will be ignored.
! = End of variant file

Data head:

[H = Start of data head

Variable table:

[0 = Start of variable table

Cartesian coordinate systems:

[K = Start of Cartesian coordinate systems

Contour elements:

All specifications in mm resp. degree

Jn = Start of contour element description of contour n

\$Em = Contour element description of contour m

KP = Descriptor of contour point

KL = Descriptor of contour line

KA = Descriptor of arc

KR = Descriptor of corner rounding

KF = Descriptor of chamfer edging

KSL = Contour split element in one line

KSA = Contour split element in a circle

Processingen:

<n : Processing n

?? : Condition sign

KM : Comment

MX : Processing is dependant from previous measurement in X

MY : Processing is dependant from previous measurement in Y

MZ : Processing is dependant from previous measurement in Z

MXF : Measurement factor X (default=1)

MYF : Measurement factor Y (default=1)

MZF : Measurement factor Z (default=1)

EN : Enabled



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3 Format of Parameter Rows

Generally the format of the parameter rows is changing Starting from version 4.0. A Parameter row inside the blocks **data head**, **Variable table** and **processings** of MPR-file consists of a Descriptor, the equals sign and the input string. The input string is stated in double high commas.

E.g.:
XA="1000-32"
YA="B/2"
DU="5"

The Format of the Parameter rows within the Blocks **contour elements** do not change. The input string is not stated in double high commas.

E.g.:
X=0.0
Y=120.0
Z=D-2

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4 Parser

Mathematic expressions will be calculated by a parser .

Following predefined marks will be recognized by the Parser :

Name/Descriptor	Meaning	Typ
_cc	counter clockwise, small arc	Constant (=0)
_cw	Clockwise, small arc	Constant (=1)
_CC	counter clockwise, big arc	Constant (=2)
_CW	Clockwise, big arc	Constant (=3)
_mirror	Mirror in X active=1, Mirror in X deactive=0	Variable
_nonmirror	Mirror in X active=0, Mirror in X deactive=1	Variable
_ymirror	Mirror in Y active=1, Mirror in Y deactive=0	Variable
_nonymirror	Mirror in Y active=0, Mirror in Y deactive=1	Variable
_BSX	Work piece lenght of processed part	Variable
_BSY	Work piece width of processed part	Variable
_BSZ	Work piece thickness	Variable
_ok	1	Constant
_no	0	Constant
_lf	Tool radius correction left	Constant (=1)
_ri	Tool radius correction right	Constant (=2)
@	Descriptor for relative measuring	Descriptor
SIN()	Sinus, Argument in degrees	Funktion
COS()	Cosinus, Argument in degrees	Funktion
TAN()	Tangens, Argument in degrees	Funktion
ARCSIN()	Arkussinus	Funktion
ARCCOS()	Arkuscosinus	Funktion
ARCTAN()	Arkustangens	Funktion
EXP()	Exponential function	Funktion
LN()	Natural Logarithmus	Funktion
SQRT()	Sqare root	Funktion
MOD()	Modulo function	Funktion
PREC()	Pre commas are not applicable	Funktion
AND	AND-links (only by conditions)	Operator
OR	OR-links (only by conditions)	Operator
NOT	NOT-links (only by conditions)	Operator
IF() THEN () ELSE ()	If condition meets, i.e. the value in IF parenthesis is not equal 0, the parameter is equal the value in THEN parenthesis. Otherwise the value in ELSE parenthesis . Is the THEN and ELSE parenthesis not filled then 0	Funktion
STANDARD	Arranges the NC-generator to use the standard value for the corresponding parameter.	Keyword



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5 Data Head

To establish the version of the MPR-file in a easy way a header is created.

The header Starts with descriptor [H and is located in the first section of the MPR-data.

[illegible]

Explanation:

VERSION: Version of MPR-file. (string parameter) Possible values:
4.0 : WoodWOP-Version 4.0

R00 - R27: Bitmap of programs. Generally, meaningfully only to use with WoodWop components. Here there can be inserted a bitmap 28 X 28. These bitmap will be showed instead of the workpiece bitmap or the component macro bitmap. The parameter R00 to R27 are optional

MAT : Machine type (HOMAG, CF-HOMAG, FK-HOMAG odře WEEKE)



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- OP : Optimating modus; The drill processes will be optimized to prozessing time.
It will be chedged if processes can be operated together.
And the sequence orof will be chedged:
= 0 => No Optimizing
= 1 => Optimizing drilling from below cociofing of the next processes.(Default).
= 2 => Best possible optimizing
- O2 : = 1 => Avoid cross drive hight by horizontal processing
O3 : = 1 => Avoid cross drive hight by processing from below
O4 : = 1 => Clockwise optimizing of vertical drillings
O5 : = 1 => Avoid cross drive hight by finish processing
- FM : Clearancemodus controls the direction of clearing the machine after processing
= 0 => No clearance
= 1 => Clear ance to rear right (default)
= 2 => Clear ance to rear left
= 3 => Clearance to front left
= 4 => Clear ance to front right
- CB : Edge grouping
ML : Max. edge lenghts which can be grouped
DN : Dynamic Mode relative (default 100)
NP : NC-program shall be generated as normal program (not mirrored)
GP : NC-program shall be generated as in X mirrored program
GY : NC-program shall be generated as in Y mirrored program
GXY : NC-program shall be generated as in X and Y mirrored program
UP : NC-program shall be generated as sub program
NE : Don't clear tool after end of program
A : Don't feed out tool after end of program (only by throughfeed table)
DW : Turn angle. NC-programme shall be turned. At present adcenterd angle : 0.90.-90
MI : Mirror. Doesn't matter if drawn mirrorwise or not. No influence of NC-generation. Determines
only the direction of the object while loading to WoodWop.
= 0 drawn unmirrored
= 1 drawn mirrored
- INCH : = 0 Measures in mm
= 1 Measures in INCH
- VIEW : Representation in WoodWOP (NOMIRROR / XMIRROR / YMIRROR / XYMIRROR)
_BSX : Boardsize in X,. These _* - parameters contain the calculated values of the work piece
description. These values are used from external evaluation programs, because an
indipendent parsen is large-scale.These informations are irrelevant for the NC-generator.
- _BSY : Boardsize in Y,.
_BSZ : Boardsize in Z,.
_FNX : Offset of finished workpiece in X
_FNY : Offset of finished workpiece in Y
_RNX : Offset of raw work piece in X
_RNY : Offset of raw work piece in Y
_RNZ : Offset of raw work piece in Z
_RX : Size of raw work piece in X
_RY : Size of raw work piece Y
- ANZ : Number of pieces to be produced (workpiece macro)
MA : Grain Direction, inactive if this parameter is missing (workpiece macro)
BES : time to apply parts to the machine (wood**Time**)
ENT : time to remove parts from the machine (wood**Time**)

6 Variable table

A variable table will be initialized by the sign '[' in the first column of a section. After that an ID-number declaration will be showed in column 2. The ID-number has to be 0. The following rows will introduced as variables untill the cartesian coordinate system, contour elements and processing macros are declared. The name of the variable starts at column 1. Its name can not exceed eight characters. The first character can't be a number but a letter. The, to the variables assigned values, follow, divided with a equals sign, in the same row. Here are also formula expressions allowed. There can only be used prespecified variables in the formulas.

After each variable a comment can be specified in the following row which closely describes the variable. These will be introduced by the descriptor KM. KM is a stringparameter. The comment is maximally 80 characters long.

E.g.:

```
[000
L="950.0"
KM="Lengths of work pieces in X"
B="600.0"
KM="Width of work pieces in Y"
Center="L/2"
KM="Mid axle"
Distance1="307.5"
KM="Distance between construction drills for hardware H3245"
```

7 Cartesian Coordinate System

The list of Cartesian coordinate systems will be produced by descriptor '[K' in the first and second column of a row. After that the coordinate systems starts with descriptor <00.

E.g.:
[K
<00 \Coordinate system\
NR="05"
XP="1500"
XF="1"
YP="0"
YF="1"
ZP="0"
ZF="1"
D1="0"
KI="0"
D2="0"
MI="0"

Explanation:

- NR : Number of Cartesian coordinate systems. 4 to 99 are the valid values at present. The coordinate systems 0 to 3 are fix and won't be saved in the MPR-file.
- XP : X-Position refers to coordinate system 0.
- XF : X-Faktor. Zoom-factor of X-axis.
- YP : Y-Position refers coordinate system 0.
- YF : Y-factor. Zoom-factor of Y-Axis.
- ZP : Z-position refers coordinate system 0.
- ZF : Z-factor. Zoom-factor of Z-Axis.
- D1 : Turn angle 1 around Z-Axis of coordinate system 0.
- KI : Tilt angle around X-Axis of new coordinate system (after turning around D1)
- D2 : Turn angle 2 around Z-Axis of new coordinate system (after turning around D1 and tilting around KI)
- MI : Mirror. Mirrored (right turned) coordinate system or wise versa.
= 0 unmirrored (left turned, resp. like in coordinate system 0)
= 1 mirrored (right turned, resp. like in coordinate system 1)

In WoodWOP are, without listed in the MPR-file, 20 Coordinate systems pre defined at the corner points of the finished work pieces:

00:	Coord. syst. XY-Level	X=0	Y=0	Z=0	Turnangle 0 X-Axis positiv
01:	Coord. syst. XY-Level	X=L	Y=0	Z=0	Turnangle 0 X-Axis negativ
02:	Coord. syst. XY-Level	X=L	Y=B	Z=0	Turnangle 180X-Axis positiv
03:	Coord. syst. XY-Level	X=0	Y=B	Z=0	Turnangle 180X-Axis negativ
A0:	Coord. syst. XZ-Level (Y+)	X=0	Y=0	Z=0	Turnangle 0 X-Axis positiv
A1:	Coord. syst. XZ-Level (Y+)	X=L	Y=0	Z=0	Turnangle 0 X-Axis negativ
A2:	Coord. syst. XZ-Level (Y+)	X=L	Y=0	Z=D	Turnangle 180X-Axis positiv
A3:	Coord. syst. XZ-Level (Y+)	X=0	Y=0	Z=D	Turnangle 180X-Axis negativ
B0:	Coord. syst. YZ-Level (X-)	X=L	Y=0	Z=0	Turnangle 0 X-Axis positiv
B1:	Coord. syst. YZ-Level (X-)	X=L	Y=B	Z=0	Turnangle 0 X-Axis negativ
B2:	Coord. syst. YZ-Level (X-)	X=L	Y=B	Z=D	Turnangle 180X-Axis positiv
B3:	Coord. syst. YZ-Level (X-)	X=L	Y=0	Z=D	Turnangle 180X-Axis negativ
C0:	Coord. syst. XZ-Level (Y-)	X=L	Y=B	Z=0	Turnangle 0 X-Axis positiv
C1:	Coord. syst. XZ-Level (Y-)	X=0	Y=B	Z=0	Turnangle 0 X-Axis negativ



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C2:	Coord. syst. XZ-Level (Y-)	X=0	Y=B	Z=D	Turnangle 180X-Axis positiv
C3:	Coord. syst. XZ-Level (Y-)	X=L	Y=B	Z=D	Turnangle 180X-Axis negativ
D0:	Coord. syst. YZ-Level (X+)	X=0	Y=B	Z=0	Turnangle 0 X-Axis positiv
D1:	Coord. syst. YZ-Level (X+)	X=0	Y=0	Z=0	Turnangle 0 X-Axis negativ
D2:	Coord. syst. YZ-Level (X+)	X=0	Y=0	Z=D	Turnangle 180X-Axis positiv
D3:	Coord. syst. YZ-Level (X+)	X=0	Y=B	Z=D	Turnangle 180X-Axis negativ

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8 Contour moves

Contour moves consists of contour elements. By contour processing from these elements will be taken reference. The elements in the MPR-File will be numbered and identified by following key letters:

KP : Contour point
KL : Contour line
KA : Contour circle arc (arc)
KR : Contour rounding (edges rounding)
KF : Contourfase
KSL : Contour split element on a line
KSA : Contour split element on a circle

The Art, in which the Element in WoodWOP-dialog are definiert (lines by end point or line by length and angle, etc.) comes of the following parameter rows. By definition of contour elements the coordinates of end points are always specified. The Start point of a contour elements is always the end point of the previous contour element. Because of this the first element of a contour move has to be always a start point. Each element can display optional to the feeding parameters the result parameters ("point value") .

8.1 Points

Start point of a contour. Key letter is **KP**. Point values are the absolute coordinates of the point in relation to the respective coordinate system and the coordinate number itself.

E.g.:
\$E0
KP
X=10.0000
Y=10.0000
Z=0.0000
KO=00
.X=10.0000
.Y=10.0000
.Z=0.0000
.KO=00

Possible parameters:

X : X-coordinate
Y : Y-coordinate
Z : Z-coordinate
KO : Number of coordinate systems

8.2 Lines

Line, defined by line dialog. Key letter is **KL**. Point values are the absolute coordinates of end points refers to the respective coordinate systems, i.e. the coordinate system of start point. The angle of the line in the X/Y level is specified with **WI**. The angle of the line to X/Y-level is **WZ**.

Angles are given in arc length
(1.57070796 = 90 degrees).

E.g.:
\$E1

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```
KL
X=1340.0000
Y=10.0000
.X=1340.0000
.Y=10.0000
.Z=0.0000
.WI=1.570796
.WZ=0.0000
```

Possible parameter:

```
X : X-coordinate of end point
Y : Y-coordinate of end point
Z : Z-coordinate of end point
L : Length
WI : Angle in X/Y-level
WZ : Angle to X/Y-level
```

8.3 Circle arcs

Circle arc, defined by circle arc dialog. Key letter is **KA**. Point values are the absolute coordinates of end pointes refers to the respective coordinate system, i.e. the coordinate system of the start point. Further will be shorten the absolut coordinates of circle midpoint with **.I, J., K.**, the increase of start and endpoint in the arc bow with **.A** and **.E**, the rotatory direction with **.DS**, the radius with **.R**, the angle of arc length in startpoint with X/Y level with **.WI**, the angle of arc length in X/Y level with **.WO** and the and the angle to X/Y level with **.WAZ**. **Angles are given In arc length.**

```
E.g.:
$E2
KA
X=1400
Y=0
DS=0
R=B/2
.X=1400.0000
.Y=0.0000
.Z=0.0000
.I=1400.0000
.J=400.0000
.DS=0
.R=400.00000
.WI=0.000000
.WO=3.141593
.WAZ=0.000000
```

Possible parameter:

```
X : X-coordinate of end points
Y : Y-coordinate of end points
Z : Z-coordinate of end points
I : Circle midpoint coordinate in X
J : Circle midpoint coordinate in Y
DS : Rotary direction
    =0 Arc <= 180 degree clockwise
    =1 Arc <= 180 degree counterclockwise
    =2 Arc > 180 degree clockwise
    =3 Arc > 180 degree counterclockwise
```



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R : Radius
WI : Start angle in X/Y-level
WO : End angle in X/Y-level
WZ : Angle to X/Y-level of connecting line from start to end point.
WAZ : Angle to X/Y-level on the arc path.

8.4 Corner rounding

Circle arcs, defined by dialog for corner rounding Key letter is **KR**. Point values are the same like by circle arcs.

E.g.:
\$E6
KR
R=50
.X=50.0000
.Y=800.0000
.Z=0.0000
.I=50.0000
.J=750.0000
.DS=0
.R=50.000000
.WI=0.785398
.WO=0.000000
.WAZ=0.000000

Possible parameter:

R : Radius

8.5 Chamfer edge

Line defined by dialog for chamfer edge. Key letter is **KF**. Point values are the same than by lines.

E.g.:
\$E3
KF
L=30
.X=1365.0000
.Y=765.0000
.Z=0.0000
.WI=0.785398
.WZ=0.000000

Possible parameter:

L : Length of chamfer
FB : Chamfer width
WI : Start angle
WO : End angle
KX : Shrinking of predecessor element.
KY : Shrinking of successor element.

8.6 Split Element

Use split element across dialog for Element splitting. Key letter are **KSL** for splitted line and **KSA** for splitted circle. Point values are the same like by lines or arcs.

E.g.:
\$E2
KSL
X=400
.X=500.000000
.Y=300.000000
.Z=0.000000
.WI=0.463648
.WZ=0.000000

Possible parameter:

X : Split point X-coordinate
Y : Split point Y-coordinate
Z : Split point Z-coordinate
AV : Distance between previous point. By negative value distance from sucession point
Relative value declaration relates to element length. E.g. AV=@0.5 means splitting in middle of element.
AK : Distance from konstruktion point. By negative value distance to sucession point.

9 Processes

Processes will be itemized in the variant file by process macros. The base macros of the workpiece creates the actual program head and is absolutely necessary. The rest processing macros are optional.

The declaration of a processing macro will be introduced by the character '<' in column 1 of a row. After that follows the ID-number and the name of the macro as comment between two '\'-Characters. The Parameter drilling from below have to follow in the next row in a specific order. All drilling from below will be drilling from below by addescriptor marks. The expression (feeding string) will be specified by double high commas. Condition parameters are only displayed when they are filled in WOODWOP.

If parameters with default are documented, they don't have to be necessary displayed in MPR-format.

9.1 Base macros

9.1.1 Werkstck : Workpiece explanation

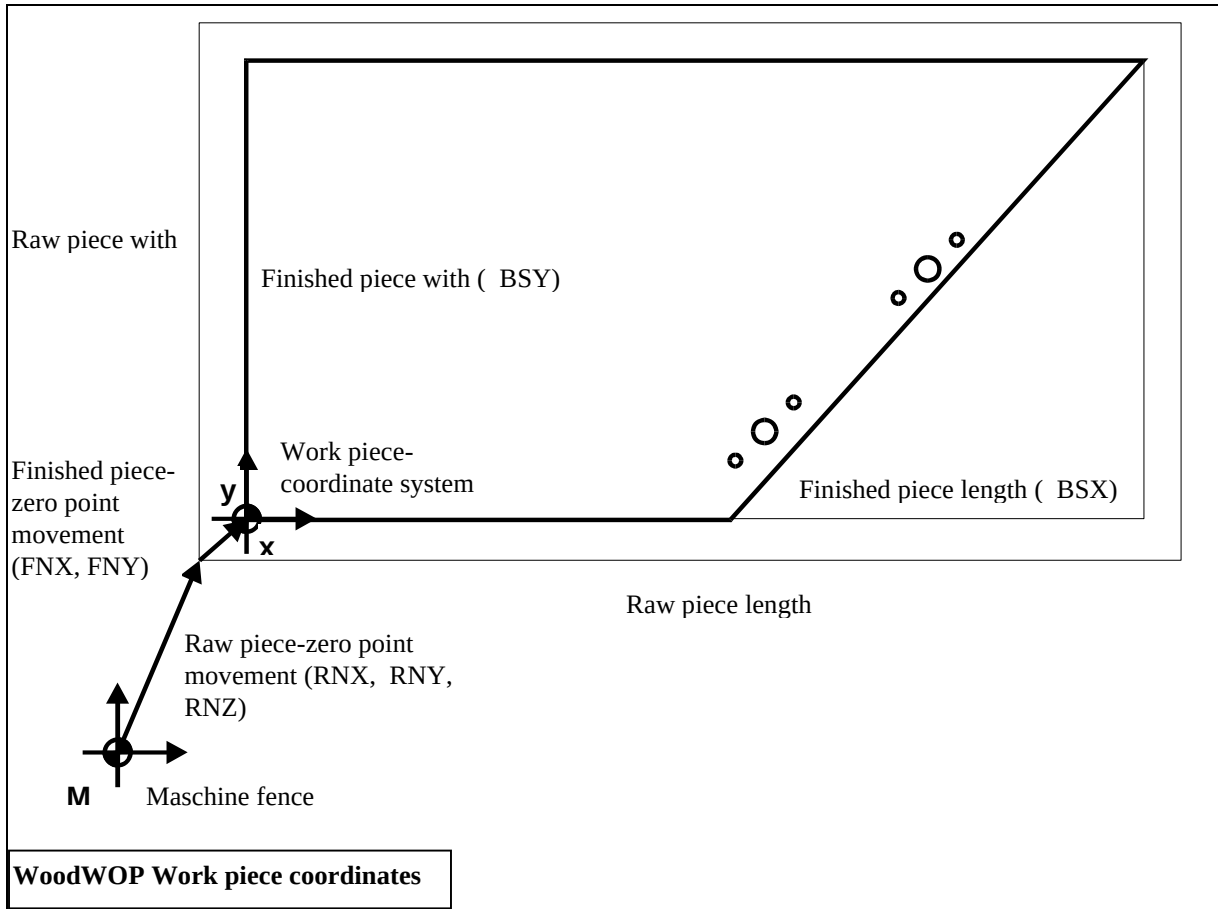
Describes the seizes of the workpiece as raw material and prefabricated part as well as the remove measure regarding to the machine fence.

ID-Number: 100

E.g.:
<100 \Werkstck\
LA="800"
BR="600"
DI="20"
AX="5"
AY="5"
RNX="100"
RNY="0"
RNZ="0"
FNX="2.5"
FNY="2.5"

Explanation:

LA : Length of finished part. (global variable _BSX)
BR : Width of finished part. (global variable _BSY)
DI : Thickness of work piece. (globale variable _BSZ)
AX : Offset in X = length of raw piece – length of finished part.
AY : Offset in Y = Width of raw piece – width of finished part.
RL : Länge of raw piece in X. Alternatively can be declared AX and AY or RL and RB.
RB : Width of raw piece in Y.
RNX : Raw piece –zero point removing regarding machine fence in X e.g. by template.
RNY : Raw piece –zero point removing regarding machine fence in Y e.g. by template
RNZ : Raw piece –zero point removing regarding machine fence in Z e.g. by higher vacuum cupss. All those processes will be moved in Z direction by this value.
FNX : Finished piece –zero point removing regarding machine fence in X e.g. by offset . All those prozesses will be moved about the value RNX+FNX in X-direction
FNY : Finished piece –zero point removing regarding machine fence in Y e.g. by offset . All those prozesses will be moved about the value RNX+FNX in Y-direction



9.1.2 Comment : Comment

Comment text, e.g. for explanation of Work piece.

ID-number: 101

E.g.:
 <101 \Comment\
 KM="Schranksseite Modell 17A9 "Erika"
 KM="Best.-Nr.: 739598"
 KM="Werkstoff: Buche hell"

Explanation:

KM : Comment row. By each CR in the comment text in WoodWOP will be inserted a new KM-insertion.
 EN : Enable (default=1)

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9.2 Processings macros

By processing macros there can be fundamental existing following parameters:

?? : Condition (*default=1*)
 EN : Enable (*default=1*)
 HP : Hood position (*default=0*)
 0 = Hood controlling automatically
 1 = Hood down position
 2 = Hood middle position (new: intermediate position 2)
 3 = Hood top position
 100= new: Hood intermediate position 1

 SP : Spindle controlling (*default=0*)
 0 = Spindle controlling automatically
 1 = Spindle left
 2 = Spindle right
 3 = Spindle using left and right
 YVE : Y-offset of second spindle, if processing has to be done with both spindles.
 (*default=0*)
 MX : Processing is depend from previous measures in X (*default=0*)
 MY : Processing is depend from previous measures in Y (*default=0*)
 MZ : Processing is depend from previous measures in Z (*default=0*)
 ASG : Exhauster
 0 = Off
 1 = On
 2 = Automatic
 RSEL : set of rules selected (1) / not selected (0) (Techauto)
 RWID : selected set of rules identifier (Techauto)
 KAT: processing category (wood**Time**)
 MNM: Macro name (localized, wood**Time**)

All others parameter will be explained in following.

9.2.1 BohrVert: Drillings vertical

ID-Number: 102

E.g.:
 <102 \BohrVert\
 XA="200"
 YA="300"
 BM="SS"
 TI="10"
 DU="5"
 \alternative zu DU: TNO="121"
 AN="10"
 MI="0"
 AB="32"
 XR="0"
 YR="1"
 S_=2
 \Optional additiv: WI="90"
 F_="STANDARD"



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KO="2"
??= "_mirror"

Explanation:

XA : Drill coordinate in X
YA : Drill coordinate in Y
BM : Drill mode (stringparameter):
 LS =0: slow-fast to depth
 SS =1: fast - fast to depth
 LSL =2: slow - fast - slow compl. across
 SSS =3: fast - fast – fast comp. Across
 LSU =6: slow - fast from the bottom
 LSLU =8: slow - fast - slow from the bottom
 All further declarations relates to self defined drill cycles.
 e.g. 001 etc..-> calls NC-sub program TVBOHR40.001.

TI : Depth of drilling
DU : Diameter of drilling (*alternative DU or TNO*)
TNO : T-number (tool number) (*alternative TNO or DU*)
LA : Lengths of hole row (*alternative LA or AN*)
AN : Number of drillings (*alternative LA or AN*)
MI : Hole row type: 0:start point- 1:center point.
AB : Raster distance of drillings
WI : hole raster direction angle in X/Y-Level (optional additiv to WI or XR and YR). The angle will be declared in degrees (e.g. 90.0). If this parameter is specified, WOODWOP will recognize that the input is given across the angle.Accordingly the input mask will be configured. The NC-generator ignores the Angle and uses only XR and YR.

XR : Hole row direction in X
YR : Hole row direction in Y
F_ : Feedrate: contains either the correct value in m/min or the key word STANDARD. By STANDARD, the feedrate of the tool datas will be used (*default*).
S_ : r.p.m.
 0: Slow
 1: Normal
 2: Fast

KO : Coordinate system:
 0: Work piece coordinate system ("left below") (*default*)
 1: Work piece coordinate system ("right below")
 2: Work piece coordinate system ("right top")
 3: Work piece coordinate system ("left top")
 4-100: Work piececoordinate system user defined

?? : condition (*default=1*)
EN : Enable (*default=1*)

9.2.2 BohrHoriz: Drilling horizontal

ID-Number: 103

E.g.:
<103 \BohrHoriz\
XA="200"
YA="300"
ZA="D/2"
DU="8"
TI="35.0"
BM="XP"
AN="3"
AB="32"

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```
F_="STANDARD"
ANA="15"
??=" _mirror"
EN="0"
```

Explanation:

XA : Drill coordinate in X
YA : Drill coordinate in Y
ZA : Drill coordinate in Z
BM : Drill mode(string parameter):
 XP =0: X-Plus, i.e. in positive X-direction
 XM =1: X-Minus, i.e. in negative X-direction
 YP =2: Y-Plus, i.e. in positive Y-direction
 YM =3: Y-Minus, i.e. in negative Y-direction
 C =4: with free C-angle
WI : Angle of drilling in X/Y-level. The angle will be declared in degrees (e.g. 45.0). These parameter ist only necessary by BM=C.
TI : Depth of drilling
DU : Diameter of drilling
AN : Number of drillings
LA : Lengths of hole row (*alternative LA or AN*)
MI : Hole row typ: 0:Start point- 1:Middle point
T_ : T-number (Tool number) (*alternative T_ or DU*)
AB : Raster distance of drillings
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD. by STANDARD the Feedrate of the tool data will be used (*default*).
ANA : Additional start distance
MD : Mode. (Extension for self defined subroutine)
KO : Coordinate system
?? : Condition (*default=1*)
EN : Enable (*default=1*)

9.2.3 BohrUniv: Drilling universal (Space drilling)

ID-number: 104

```
E.g.:
<104 \BohrUniv\
XA="500"
YA="300"
ZA="17"
CA="45"
WI="9"
DU="8"
TI="20"
AN="3"
AB="32"
F_="STANDARD"
```

Explanation:

XA : Drill coordinate in X (first Drilling) (*alternative XA or XM*)
YA : Drill coordinate in Y (first Drilling) (*alternative YA or YM*)
XM : Drill coordinate in X (center hole row) (*alternative XA or XM*)
YM : Drill coordinate in Y(center hole row) (*alternative YA or YM*)
ZA : Drill coordinate in Z
CA : Turn angle C-Axis in of X-Y-Level, declared in degrees.
WI : Swing angle of space drill unite, declared in degrees.



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DU : Diameter of drilling
TI : Depth of drilling
AN : Number of drillings
LA : Lengths of hole row (*alternative LA or AN*)
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
by STANDARD the Feedrate of the tool data will be used (*default*).
KO : Coordinate system
?? : Condition (*default=1*)
EN : Enable (*default=1*)

9.2.4 UfluBohr: drilling from below

ID-number: 131

E.g.:
<131 \UfluBohr\
XA="0"
YA="0"
DU="8"
WI="0"
TI="0"
AB="10"
F_="STANDARD"
EN="0"

Explanation:

XA : Coordinate in X
YA : Coordinate in Y
DU : Drilling diameter
WI : C-angle
TI : Drill depth
AB : Distance of dive in / dive out of drilling holes in direction of C-angles
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
By STANDARD the feedrate of tool data will be used (*default*).
MD : Modus. (extension of self defined start makros)
KO : Coordinate system
ZA : Z-Reference value for the depth
?? : Condition (*default=1*)
EN : Enable (*default=1*)

9.2.5 Grooving: Grooving and Sawing

ID-number: 109

E.g.:
<109 \grooveen\
XA="0"
YA="200"
XE="L"
YE="400"
NB="5"
RK="WRKL"
EM="MOD0"
TV="5"
MV="GL"
TI="67"
MN="GL"



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OP="1"
T_="140"
F_="STANDARD"
EN="0"

Explanation:

XA : Start coordinate in X
YA : Start coordinate in Y
XE : End coordinate in X
YE : End coordinate in Y
WI : Angle, declaration only necessary by AN = 1 or 2
AN : Selection of feeding in way. (default=0)
0 = X ENDE_Y ENDE
1 = X ENDE_ANGLE
2 = Y ENDE_ANGLE
NB : Groove width
RK : Radius correction:
NOWRK =0: Tool runs on programmed line
WRKL =1: Toolruns left of programmed line
WRKR =2: Tool runs right of programmed line
EM : Insert mode
MOD0 =0: Groove has at start- and end-coordinates full depth
MOD1 =1: Groove starts at start- and end-coordinates
MOD2 =2: Start and end-coordinates will be calculated with safety distance.
Suitable for grooving through entire work piece.
FM : Feed mode: self defined insertion-resp. start mode (Stringparameter). (alternative declare FM
or
EM)
TV : depth
MV : Mode scoring
GL =0: downcut scoring
GGL =1: conventional scoring
TI : Groove depth
MN : Mode grooving
GL =0: downcut grooving
GGL =1: conventional grooving
OP : Optimizing
0: optimizing off. Free movements by wide groove.
1: optimizing on. Return drive by processing by wide groove.
XY : Deliver factor
T_ : Tool number, unit
F_ : Feedrate: contains either the concrete value in m/min or the keyword STANDARD.
By STANDARD the feedrate will be taken from the tool datas (default).
KO : Coordinate system
?? : Condition (default=1)
EN : Enable (default=1)

9.2.6 groove_R: Angled Grooving

ID-number: 124

E.g.:
<124 \groove_R\
XA="0"
YA="0"
XE="0"
YE="0"

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```
ZA="0"
RK="WRKL"
WI="45"
EM="MOD0"
TV="0"
MV="GL"
TI="0"
MN="GL"
```

Explanation:

XA : Start coordinate in X
 YA : Start coordinate in Y
 XE : End coordinate in X
 YE : End coordinate in Y
 ZA : Start coordinate in Z
 RK : Radius correction:
 NOWRK =0: Tool runs on programmed line
 WRKL =1: Toolruns left of programmed line
 WRKR =2: Tool runs right of programmed line
 EM : Insert mode
 MOD0 =0: Groove has at start- and end-coordinates full depth
 MOD1 =1: Groove starts at start- and end-coordinates
 MOD2 =2: Start and end-coordinates will be calculated with safety distance.
 Suitable for grooving through entire work piece.
 FM : Feed mode: self defined insertion-resp. start mode (string parameter). (alternative declare FM or EM)
 TV : Scoring depth (declaration alternative to scoring Z value)
 VZ : Z-value for the scoring (declaration alternative to scoring depth)
 MV : Mode scoring
 GL =0: downcut scoring
 GGL =1: conventional scoring
 XY : Art of declaration of XY-coordinates
 0: XY-declaration to Z-start-level
 1: XY-declarations to Z-zero-level
 TI : Groove depth (declaration alternative to Z-value)
 Z_ : Z-value of groove (declarations alternative to depth)
 MN : Modus grooving
 GL =0: downcut grooving
 GGL =1: conventional grooving
 T_ : Toolnumber, unite
 F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
 By STANDARD theFeedrate of the Tooldatas will be used (default).
 XY : Height of XY-value
 0: XY-Value on Z-Level
 1: XY- Value on Z-0-Level
 TI : Depth of the groove (Instead of Z-Value)
 KO : Coordinate system
 ?? : condition (default=1)
 EN : Enable (default=1)

9.2.7 Polygonzug: Polygon milling

ID-number: 119

```
E.g.:
<119 \Polygonzug\
NM="_gk.ply"
```

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```
XA="0"
YA="0"
BR="46.6"
HE="100"
WI="0"
TI="3"
WZ="101"
RK="NOWRK"
F_="5"
```

Explanation:

NM : Name of Polygon. Reference to a polygon file *.ply.

XA : Center coordinate in X

YA : Center coordinate in Y

BR : Width (Dimension in X)

HE : Height (Dimension in Y)

WI : Turn angle

TI : Depth

WZ : Tool number

RK : Radius correction:

NOWRK =0: Tool runs on programmed line

WRKL =1: Toolruns left of programmed line

WRKR =2: Tool runs right of programmed line

8.2.8 G F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
By STANDARD theFeedrate of the Tooldatas will be used (default).

KO : Coordinate system

?? : condition (default=1)

EN : Enable (default=1)

LOCAL : Polygon Milling can be saved locally (default=0)

STARTLOCAL : start-flag for local Polygon Milling (if LOCAL=1)

ENDLOCAL : end-flag for local Polygon Milling

HU : Tracing stroke for unit with mechanical tracing

MT : mechanical tracing on(1) or off(0)

Note: if mechanical tracing is active, depth(TI) is disabled.

Tracing stroke is only enabled if mechanical tracing(HU) is active

9.2.8 Graphic Comment

ID-number: 152

```
E.g.:
<152 \Grafischer Comment\
NM="BOHRKOPF.PLY"
MD="1"
XA="0"
YA="0"
BR="36.0"
HE="38.9"
WI="45"
KO="01"
ROT="0"
GRUEN="0"
BLAU="255"
STYLE="0"
```

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Explanation:

NM : Name of Polygon. Reference to a polygon file *.ply.
 XA : Center coordinate in X
 YA : Center coordinate in Y
 BR : Width (Dimension in X)
 HE : Height (Dimension in Y)
 WI : Turn angle
 ROT : Red-Part of the display color
 GRUEN: Green- Part of the display color
 BLAU : Blue- Part of the display color
 STYLE: Linestyle for display (0:SOLIDPEN, 1: DASH, 2: DOT, 3: DASHDOT, 4: DASHDOTDOT,5: NONE)
 WIDTH: Line width in pixel (*default=1*)
 KO : Coordinate system
 ?? : condition (*default=1*)
 EN : Enable (*default=1*)
 LOCAL : Graphic Comment can be saved locally (*default=0*)
 STARTLOCAL : start-flag for local Graphic Comment (if LOCAL=1)
 ENDLOCAL : end-flag for local Graphic Comment

9.2.9 Pocket: Vertical pocket milling

ID-number: 112

E.g.:
 <112 \Tasche\
 XA="0"
 YA="0"
 LA="120"
 BR="20"
 RD="5"
 WI="0"
 TI="2"
 ZT="0"
 XY="80"
 DS="1"
 T_="101"
 F_="STANDARD"

Explanation:

XA : Center coordinate in X
 YA : Center coordinate in Y
 LA : Length (dimension in X)
 BR : Width (dimension in Y)
 RD : Edges radius
 WI : Turn angle
 TI : Depth
 ZA : Z-Reference value for the depth
 ZT : Deliver depth
 XY : Delivering in XY-level
 T_ : Tool number
 F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
 By STANDARD the Feedrate of the Tooldata will be used (*default*).
 DS : Turning direction: 0 = clockwise, 1= counter clockwise
 OSZVS : feed for oscillating routing



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OSZI : enable oscillating routing yes/no (default : 0)
KO : Coordinate system
?? : Condition (default=1)
EN : Enable (default=1)

9.2.10 HTasc: Horizontal Pocket Routing

ID-number: 123

E.g.:
<123 \HTasc\
XA="0"
YA="120"
ZA="9"
LA="120"
BR="40"
RD="4"
WI="90"
TI="20"
ZT="0"
AB="10"
T_="101"
F_="STANDARD"
DS="1"

Explanation:

XA : Center coordinate in X
YA : Center coordinate in Y
ZA : Center coordinate in Z
LA : Length (dimension in XY- level vertical to Tool)
BR : Width (dimension in Z)
RD : Edge radius
WI : Turn angle (in XY-level)
TI : Depth
ZT : Deliver depth
MD : Mode. (extension of self defined start macros)
AB : Start distance
T_ : Tool number
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
By STANDARD the Feedrate of the Tooldata will be used (default).
DS : Turning direction: 0 = clockwise, 1= counter clockwise
KO : Coordinate system
?? : Condition (default=1)
EN : Enable (default=1)

9.2.11 FreiFormTasche (Free form pocket)

ID-number: 181

E.g.:
<181 \FreiFormTasche\
EA="1:0"
AD="0"
AZ="0"
ZU="0.5"
TI="0"



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ZT="0"
XY="80"
T_="101"
F_="5"
DS="1"
HP="0"
SP="0"
YVE="0"
WW="1,3,401,403"
ASG="2"
MX="0"
MY="0"
MZ="0"
MXF="1"
MYF="1"
MZF="1"

Erklärung:

EA : Number of start elements
AD : Distance for last path to clear (normally the outermost contour)
AZ : Start distance in Z
ZU : Infeed speed for sensitive workpieces
TI : Depth
ZT : Step depth
XY : Infeed in XY
T_ : Tool number
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
By STANDARD the Feedrate of the Tooldata will be used (default).
DS : direction of rotation: 0 = clockwise, 1= counter clockwise
HP : Hood position (default=0)
0 = Hood controlling automatically
1 = Hood down position
2 = Hood middle position
3 = Hood top position
100= New: Hood inclined position 1
SP : Spindle controlling (default=0)
0 = Spindle controlling automatically
1 = Spindle left
2 = Spindle right
3 = Spindle using left and right
YVE : Y-offset of second spindle, if processing has to be done with both spindles.
(default=0)
WW : default tool numbers suitable for this macro
ASG : Exhauster
0 = Off
1 = On
2 = Automatic
MX : Processing is depend from previous measures in X (default=0)
MY : Processing is depend from previous measures in Y (default=0)
MZ : Processing is depend from previous measures in Z (default=0)
MXF : Measuring factor in X
MYF : Measuring factor in Y
MZF : Measuring factor in Z
HU : Tracing stroke for unit with mechanical tracing
MT : mechanical tracing on(1) or off(0)
Note: if mechanical tracing is active, depth, step depth and feed adjustment are disabled.



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OSZVS : *Tracing stroke is only enabled if mechanical tracing is active*
feed for oscillating routing
OSZI : enable oscillating routing yes/no (default : 0)

9.2.12 VTasche: Vector Pocket Routing

ID-number: 141

E.g.:
<141 \VTasche\
XA="100"
YA="200"
LA="120"
BR="50"
RD="10"
WI="45"
TI="10"
ZT="10"
XY="80"
T_="225"
F_="5"
DS="1"
A_="0"
C_="90"

Explanation:

XA : Center coordinate in X
YA : Center coordinate in Y
LA : Length (dimension in XY- level vertical to Tool)
BR : Width
RD : Edge radius
WI : Turn angle (in XY-level)
TI : Depth
ZT : Deliver depth
MD : Modus. (extension of self defined start macros)
XY : Delivery
T_ : Tool number
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
By STANDARD the Feedrate of the Tooldatas will be used (default).
DS : Turning direction: 0 = clockwise, 1= counter clockwise
A_ : A-Angle
C_ : C-Angle
KO : Coordinate system
?? : Condition (default=1)
EN : Enable (default=1)

9.2.13 UflurTasche: Pocketing from below

ID-number: 151

E.g.:
<151 \UflurTasche\



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```
XA="0"  
YA="0"  
LA="120"  
BR="20"  
RD="5"  
WI="0"  
AB="30"  
AWI="90"  
TI="2"  
ZT="0"  
XY="80"  
DS="1"  
T_="125"  
F_="STANDARD"
```

Explanation:

XA : Center coordinate in X
YA : Center coordinate in Y
LA : Length (dimension in XY- level vertical to Tool)
BR : Width
RD : Edge radius
WI : Turn angle (in XY-level)
AWI : Angle of unite
TI : Depth
ZA : Z-Reference value for the depth
ZT : Deliver depth
AB : Start distance
AM : Inserting mode
 0: vertical diving
 1: dive in the move
MD : Modus. (extension of self defined start macros)
XY : Delivery
T_ : Tool number
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
 By STANDARD theFeedrate of the Tooldatas will be used (default).
DS : Turning direction: 0 = clockwise, 1= counter clockwise
KO : Coordinate system
?? : Condition (default=1)
EN : Enable (default=1)

9.2.14 NC-Code: Free NC-Code

ID-Number: 116

```
E.g.:  
<116 \NC-Code\  
PA="(***** Freier NC-Code *****)"  
UPG="1"  
NM=""  
TA="1"  
PR="1"  
XA="0.0"  
YA="0.0"  
ZA="0.0"
```



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Explanation:

PA : NC-row (can occur more than once)
UPG : Inserting drilling from below yes/no 1/0
NM : Drilling from below file names
TA : Tool tcleaning yes/no 1/0
PR : parse NC-row values yes/no 1/0
XA : Insert coordinate in X
YA : Insert coordinate in Y
ZA : Insert coordinate in Z
KO : Coordinate system
?? : condition (default=1)
EN : Enable (default=1)

9.2.15 NC-Stop: NC-Stop-Makro

ID-Number: 117

E.g.:

```
<117 \NCStop\  
VL="1"  
XA="34"  
YA="45"  
_Y="1"  
XV="0"  
YV="0"
```

Explanation:

XA : X-position (drive away position)
YA : Y-position
XV : X-offset (after release vacuum land tighten again)
YV : Y-offset
VL : Release vacuum 1/0
KM : Comment
MD : Mode. (extension of self defined start macro)
_Y : Declare Y-coordinate 1/0
KO : Coordinate system
ZR : display message at realtime
WT : time to wait
?? : condition (default=1)
EN : Enable (default=1)

9.2.16 Universal Macro

ID-number: 153

E.g.:

```
<153 \Universal Makro\  
NM=""  
XA="100"  
YA="20"  
ZA="10"  
WI="25"  
WZ="77"  
S_="STANDARD"  
F_="STANDARD"  
P1="0"
```

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P2="0"
P3="0"
P4="0"

Explanation:

XA : Coordinate in X
YA : Coordinate in Y
ZA : Coordinate in Z
WI : Angle
WZ : Tool
F_ : Feedrate: contains either the concrete value in m/min or the key word STANDARD.
By STANDARD theFeedrate of the Tooldatas will be used (*default*).
S_ : Revolution per minute: contains either the concrete value in m/min or the key word STANDARD.
By STANDARD the revolution will be used (*default*).
P1 : Free parameter
P2 : Free parameter
P3 : Free parameter
P4 : Free parameter
KO : Coordinate system
?? : condition (*default=1*)
EN : Enable (*default=1*)
LOCAL : Subroutine can be saved locally (*default=0*)
STARTLOCAL : start-flag for local subroutine (if LOCAL=1)
ENDLOCAL : end-flag for local subroutine
SM : Revolution mode
0: revolutions in percent
1: in RPM
PA : NC-row (can occur more than once)
PR : parse NC-row values yes/no 1/0 (*default = 1*)

9.2.17 Messen: Measure

ID-number: 129

E.g.:
<129 \Messoriz\
XA="0"
YA="0"
ZA="0"
TI="20"
MD="X"
MP="0"

Explanation

XA : X-start position
YA : Y- start position
ZA : Z- start position
TI : depth
MD : Measure direktion "X", "Y" or "Z"
MP : Measure point
PO : Position (0: at cross 1: at tappet)
T_ : Tool
AB : Distance of dive movement from start position (door rabbit measuring)



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KO : Coordinate system
?? : condition (*default=1*)
EN : Enable (*default=1*)

9.2.18 Probing Actual Location of Workpiece

ID-number: 132

E.g.:
<132 \LageVermessen\
X1="0"
Y1="B/2"
Z1="10"
X2="100"
Y2="0"
Z2="10"
X3="L-100"
Z3="10"
MM="VR"
WI="90"
WZ="39"
VR="0"

Explanation:

X1 : Measure point 1 in X
Y1 : Measure point 1 in Y
Z1 : Measure point 1 in Z
X2 : Measure point 2 in X
Y2 : Measure point 2 in Y
Z2 : Measure point 2 in Z
X3 : Measure point 3 in X
Y3 : Measure point 3 in Y
Z3 : Measure point 3 in Z
MM : Measure mode (string parameter):
 0: HL (back left, triple measuring)
 1: HR (back right, triple measuring)
 2: VL (front left, triple measuring)
 3: VR (front right, triple measuring)
 4: H (from back single measuring)
 5: V (from front single measuring)
 6: L (from left single measuring)
 7: R (from right single measuring)
WI : Included Angle
WZ : Tool
VR : Measuring on raw piece edge
PO : Position (0: at cross 1: at tappet)
?? : condition (*default=1*)
EN : Enable (*default=1*)

9.2.19 Block: Block macro

Allows the combining of several processes to a block. All processes in a block

- Will be shifted by the offset in the block. (excluded contour processing. These relates to the non changed contour).
- Get the coordinate system of the block.

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- Will be calculated with the conditions of the block. (AND-link).
 - And will be switched with the block. (EN).
- Blocks can be saved and loaded . Files with the saved blocks have the extension **.blk**.

ID-number: 121

E.g.:

```
<121 \Block\
XP="0.0"
YP="0.0"
ZP="0.0"
NM="topfband.blk"
DP="3"
```

Explanation:

XP : Position (Offset) in X
YP : Position (Offset) in Y
ZP : Position (Offset) in Z
NM : Name of blocks (filename if block was loaded from a file)
DP : Dependent, i.e. Number of depended makros in a block
KO : Coordinate system
?? : condition (default=1)
EN : Enable (default=1)
CS : CNC-Loop (default=0)

9.2.20 Komponente: Component Macro

Refers to a separate WoodWOP-program (MPR-file) which will be included in the main document.
All processes and contourelements in the MPR-file

- Will be shifted by the offset in the block.
- Get the coordinate system of the block.
- Will be calculated with the conditions of the block. (AND-link).
- And will be switched with the block. (EN).

Components can include a refer to an extern MPR-file or to a embed MPR-file. If the MPR-file is embeded, follows after the finish characters of the main document („!"-Characters) the komplette MPR-file of the components. The component file will practically be „appendet“ to a main program. MPR-files which are used in the component macros have the extension **.mpr** , but are located for usual in a separat list (./ML4).

ID-number: 139

E.g.:

```
<139 \Komponente\
IN="topfband.mpr"
XA="0.0"
YA="0.0"
ZA="0.0"
EM="0"
VA="Turnwink 0"
VA="Durchm1 17"
VA="Drilling 1"
VA="Durchm2 12"
VA="depth 16"
```

Explanation:

IN : Name of Include-file
XA : Position (Offset) in X



PRODUCT INFORMATION

Subject : 4.2.7

Description
woodWOP- file

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YA : Position (Offset) in Y
ZA : Position (Offset) in Z
EM : Embed mode
0 = to Include-files will be referred externaly.
1 = Include-File is imbeded internaly.
PR : Privat, protected (optional) (*default=0*)
0 = variable values changeable;_BSX, _BSY, ..., and coordinate systems from
the main document counts also in the macros of the include-file
1 = variable values are inserted as fix in the variable list of the include-file ; _BSX, _BSY, ...,
and coordinate systems from the main document counts not in the macros of the include-file
VA : Variable expressions of the components, as many as. As variables will be used the entrys from
the variable table of the include-file. The variable value will be seperated by free
characters from
The variable name. "<variable name> <Space> <parameter entry>"
KO : Coordinate system
?? : Condition (*default=1*)
EN : Enable (*default=1*)